

We can see that the transient response and steady state response depends on the values of the poles of the second order transfer function $-\zeta\omega_n \pm \omega_n\sqrt{\zeta^2 - 1}$ given in Equation (23). It is already explained that if the system is overdamped, the poles are negative real numbers given by

$$-\zeta\omega_n - \omega_n\sqrt{\zeta^2 - 1} \quad \text{and} \quad -\zeta\omega_n + \omega_n\sqrt{\zeta^2 - 1}$$

If it is critically damped, we have a double pole on the real axis at $-\zeta\omega_n$. For underdamped system, the poles are complex conjugate with negative real part given by $-\zeta\omega_n \pm j\omega_n\sqrt{1 - \zeta^2}$

Let us consider a transfer function $G(s) = \frac{25}{s^2 + 2s + 25}$. The poles are obtained from the characteristic equation $s^2 + 2s + 25 = 0$, the roots of which are $s = -1 \pm j4.9$. (We can also find the poles by knowing the values of $\zeta = 0.2$ and $\omega_n = 5$ from the transfer function so that $-\zeta\omega_n = -1$ and $\omega_d = \omega_n\sqrt{1 - \zeta^2} = 4.9$)

We can plot these poles on the s-plane which is shown in fig. 10 a.

$$\text{The distance } OP_1 = \sqrt{(-\zeta\omega_n)^2 + (\omega_d)^2} = \sqrt{(\zeta\omega_n)^2 + (\omega_n\sqrt{1 - \zeta^2})^2} = \sqrt{\zeta^2\omega_n^2 + \omega_n^2(1 - \zeta^2)} = \omega_n$$

The angle θ can be found out using any of the following relations

$$\tan \theta = \frac{\omega_n\sqrt{1 - \zeta^2}}{\zeta\omega_n} = \frac{\sqrt{1 - \zeta^2}}{\zeta} \quad \dots(32a)$$

$$\cos \theta = \frac{\zeta\omega_n}{\omega_n} = \zeta \quad \dots(32b)$$

$$\sin \theta = \frac{\omega_n\sqrt{1 - \zeta^2}}{\omega_n} = \sqrt{1 - \zeta^2} \quad \dots(32c)$$

Where ever be the pole location, the distance between the pole and the origin is always ω_n . If we fix ω_n and change the value of ζ , the pole location moves along the circular path as shown by the blue * in fig. 10 b.

If we fix ζ and change the value of ω_n , the location of the pole moves along the line inclined at an angle $\theta = \cos^{-1} \zeta$ as shown by the brown o in fig. 10 b.

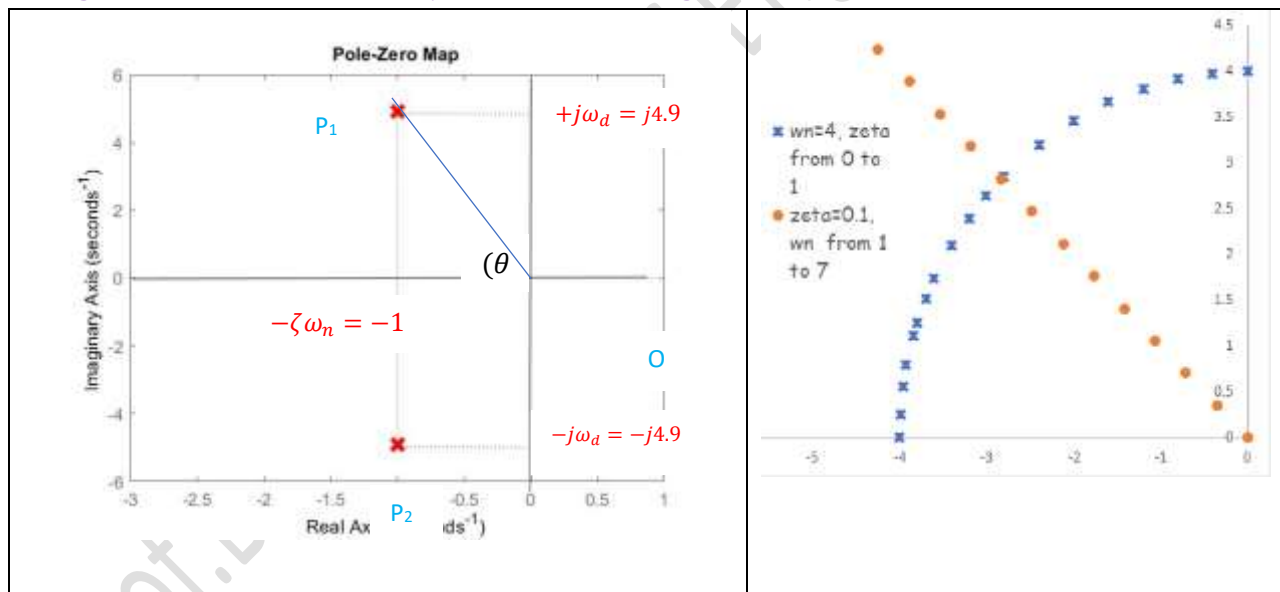


Fig.10 a. The pole-zero plot of $G(s) = \frac{25}{s^2 + 2s + 25}$

b. movement of the pole location with ζ, ω_n

You might have observed the influence of ζ and ω_n (or rather the location of the poles) on the transient response while doing #Exercise 16. The figure 11 shows the influence of damping factor and natural frequency on the step response. What are your inferences?

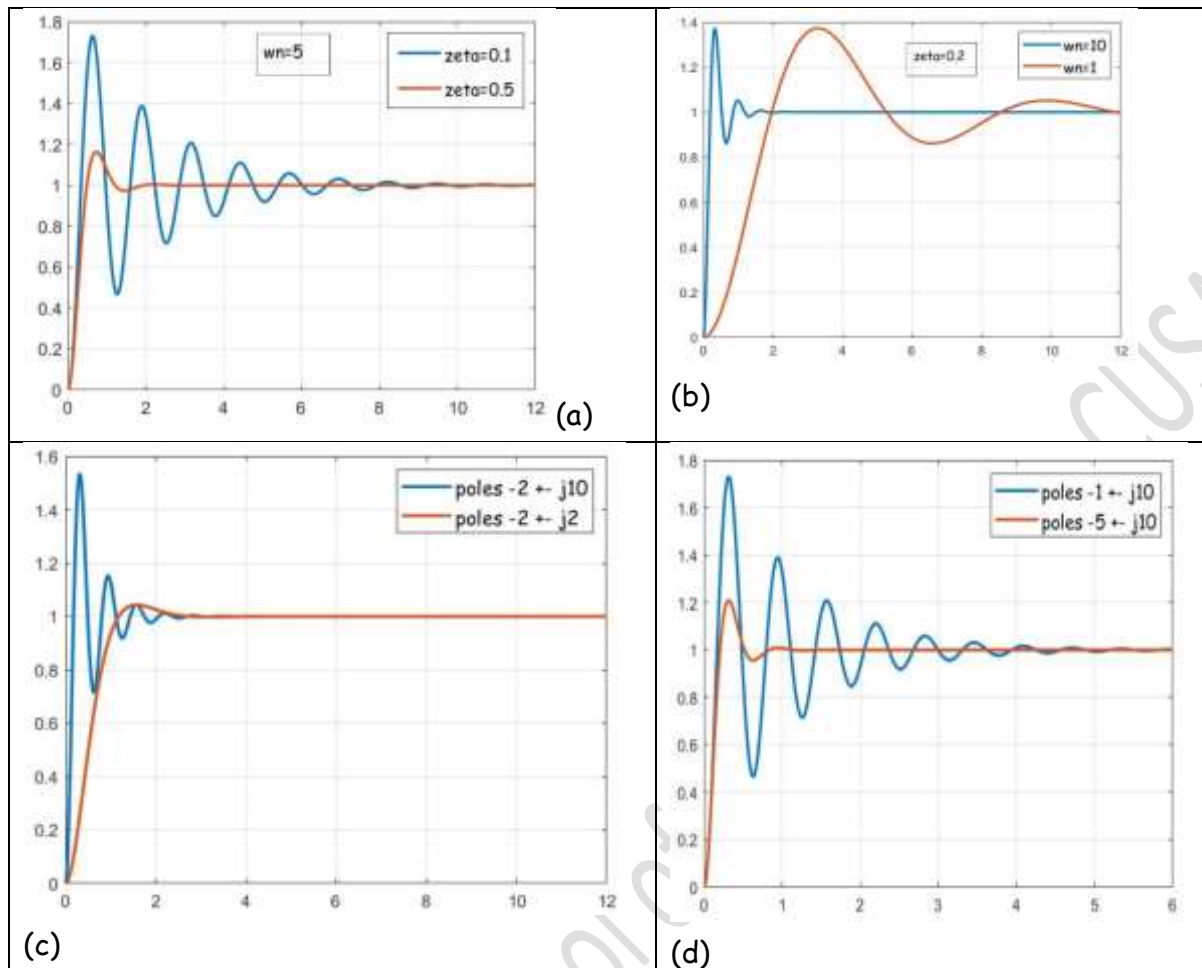


Fig.11. Comparison of the step responses for different values of damping factor and frequency

Please remember that the type of transient response is determined by the closed loop poles while the shape of the transient response is primarily determined by the closed loop zeroes.

We have already seen that depending on the location of the poles, the transient response can be undamped, underdamped, critically damped, or overdamped. Now see the influence of location of zero on the transient response.

Look at fig. 12. They represent the unit step responses of

i) $G(s) = \frac{25}{s^2 + 5s + 25}$ (No zero)

ii) $G(s) = \frac{25s}{s^2 + 5s + 25}$ (zero at $z=0$)

iii) $G(s) = \frac{25(s+1)}{s^2 + 5s + 25}$ (zero at $z = -1$)

iv) $G(s) = \frac{25(s+4)}{s^2 + 5s + 25}$ ((zero at $z = -4$)

Poles remain the same.

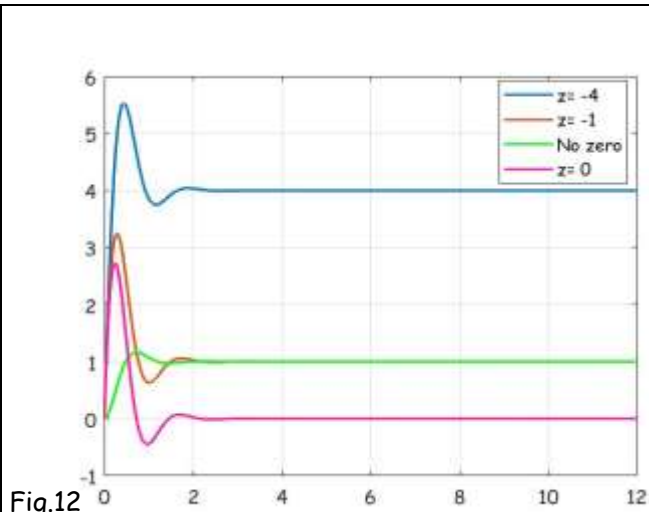


Fig.12

The transient behavior of an underdamped system can be specified by rise time, peak time, settling time, and maximum percentage overshoot (Refer Figure 9).

Rise time (t_r): It is the time taken by the system to reach the desired output for the first time. When $t = t_r$, $x(t) = 1$. Therefore, Equation (31) can be written as

$$1 = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t_r} \sin(\omega_d t_r + \theta)$$

$$\frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t_r} \sin(\omega_d t_r + \theta) = 0$$

$$\sin(\omega_d t_r + \theta) = 0$$

$$\omega_d t_r + \theta = 0, \pi, 2\pi \dots \dots \text{(If we consider 0, we get negative time. Hence we take } \pi \text{)}$$

$$t_r = \frac{\pi - \theta}{\omega_d} \dots \dots \dots (33a) \quad \text{where } \theta = \cos^{-1} \zeta \text{ and } \omega_d = \omega_n \sqrt{1 - \zeta^2}$$

Peak Time (t_p): It is the time taken by the system to reach the maximum (peak) value of the output. When $x(t)$ is maximum, $\frac{d}{dt} x(t) = 0$. Differentiating Equation (31) with respect to time

$$\frac{d}{dt} x(t) = -\frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t_p} \cos(\omega_d t_p + \theta) \omega_d + \sin(\omega_d t_p + \theta) \left\{ -\frac{1}{\sqrt{1-\zeta^2}} (-\zeta\omega_n) e^{-\zeta\omega_n t_p} \right\} = 0$$

$$\cos(\omega_d t_p + \theta) \omega_d = \sin(\omega_d t_p + \theta) \zeta \omega_n$$

$$\frac{\sin(\omega_d t_p + \theta)}{\cos(\omega_d t_p + \theta)} = \frac{\omega_d}{\zeta \omega_n} = \frac{\sqrt{1-\zeta^2}}{\zeta} = \tan \theta \quad \text{(from 32a)}$$

$$\tan(\omega_d t_p + \theta) = \tan \theta = \tan(\pi + \theta)$$

$$\omega_d t_p + \theta = \pi + \theta$$

$$t_p = \frac{\pi}{\omega_d} \dots \dots \dots (33b)$$

Maximum overshoot: It is the maximum value reached by the system response above the desired output. When $t = t_p$, $x(t_p)$ is maximum

$$x(t_p) = 1 - \frac{1}{\sqrt{1-\zeta^2}} (e^{-\zeta\omega_n t_p}) \sin(\omega_d t_p + \theta) = 1 - \frac{1}{\sqrt{1-\zeta^2}} \left(e^{-\zeta\omega_n \frac{\pi}{\omega_d}} \right) \sin\left(\omega_d \frac{\pi}{\omega_d} + \theta\right)$$

$$= 1 - \frac{1}{\sqrt{1-\zeta^2}} \left(e^{-\zeta\omega_n \frac{\pi}{\omega_n \sqrt{1-\zeta^2}}} \right) \sin(\pi + \theta) = 1 + \frac{1}{\sqrt{1-\zeta^2}} \left(e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \right) \sin \theta$$

$$= 1 + \frac{1}{\sqrt{1-\zeta^2}} \left(e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \right) \sqrt{1-\zeta^2} = 1 + e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \quad \text{(from 32 c)}$$

$$\text{Maximum overshoot} = x(t_p) - \text{desired output} = \left(1 + e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \right) - 1 = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$$

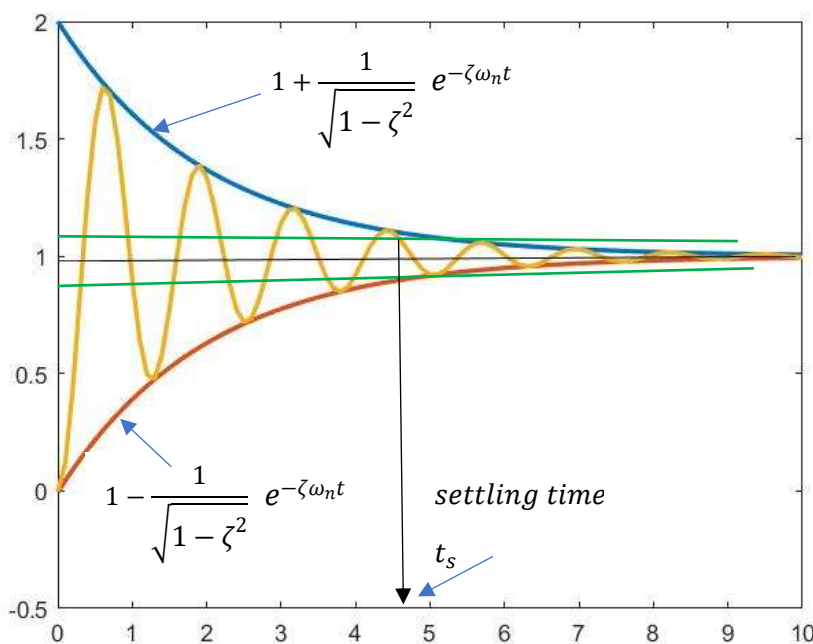
$$\text{Max. overshoot} = e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \dots \dots \dots (33c) \quad \text{(it is independent of } \omega_n \text{)}$$

Settling time (t_s): The amplitude of the underdamped response will be always between the envelopes $1 \pm \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t}$ as shown in figure 11. Time constant of these envelopes is $\frac{1}{\zeta\omega_n}$. The

speed of response depends on the value of this time constant. The settling time is defined as the time required for the system to settle within a certain percentage (usually $\pm 2\%$) of the desired output. This $\pm 2\%$ band is shown by the green line in fig.11. Settling time is the time taken by the system after which the output completely remains within the band. This happens approximately when $e^{-\zeta\omega_n t_s} < 0.02$

Or $t_s \cong \frac{4}{\zeta\omega_n}$ (33d)

Having learnt the response of second order systems, we can appreciate the influence of poles on the system response. The poles which are far away from the origin will not have much influence on the transient response. The complex conjugate poles near the origin of the s-plane relative to the other poles dominate the transient response and are called **dominant poles**. If we have a real pole and complex poles, if the ratio of real pole to the real part of the complex pole exceeds 5, then the complex poles are considered as dominant. Even if the system is of higher order, if there are dominant poles the system can be reduced to second order.



****Example**

For a negative unity feed back system with forward transfer function $G(s) = \frac{25}{s(s+7)}$ find the rise time, peak time, 2% settling time, and maximum percentage overshoot.

Given $G(s) = \frac{25}{s(s+7)}$ $H(s)=1$

Transfer function of the system = $\frac{Y(s)}{X(s)} = \frac{G(s)}{1+G(s)H(s)} = \frac{\frac{25}{s(s+7)}}{1+\frac{25}{s(s+7)}} = \frac{25}{s^2+7s+25}$

Comparing this with the standard form $\frac{\omega_n^2}{s^2+2\zeta\omega_n s+\omega_n^2}$

$\omega_n^2 = 25$, and $2\zeta\omega_n = 7$

Hence $\omega_n = 5 \text{ rad/s}$, and $\zeta = \frac{7}{2\omega_n} = 0.7$ (The system is underdamped)

Rise time $t_r = \frac{\pi-\theta}{\omega_d}$ where $\theta = \cos^{-1} \zeta = 45.6^\circ = 0.79 \text{ rad}$

$\omega_d = \omega_n \sqrt{1-\zeta^2} = 3.57 \text{ rad/s}$

$t_r = \frac{\pi-0.79}{3.57} = 0.659 \text{ s}$

$t_p = \frac{\pi}{\omega_d} = 0.88 \text{ s}$

$t_s \cong \frac{4}{\zeta\omega_n} = 1.14 \text{ s}$

% max overshoot = $e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} \times 100 = 4.5\%$

****Example**

A second order control system has the following specifications. Find the required values of the natural frequency and damping ratio of the system. What will be its rise time & peak time?

1. Maximum percentage overshoot permitted is 5%.
2. 2% settling time must be less than 4

Given % overshoot = 5% ($\omega_d = \omega_n \sqrt{1-\zeta^2} = 1.05 \text{ rad/s}$)

($\theta = \cos^{-1} \zeta = 46.4^\circ = 0.809 \text{ rad}$)

$e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}} = 0.05 \gg \gg -\frac{\pi\zeta}{\sqrt{1-\zeta^2}} = \ln 0.05 = -2.995$
 $\pi^2 \zeta^2 = 8.974(1-\zeta^2) \gg \gg \zeta = 0.69$

Settling time $t_s = 4$

$\frac{4}{\zeta\omega_n} = 4 \gg \gg \omega_n = 1.45 \text{ rad/s}$

Peak time $t_p = \frac{\pi}{\omega_d} = 2.99 \text{ s}$

Rise time $t_r = \frac{\pi-\theta}{\omega_d} = \frac{\pi-0.809}{1.05} = 2.22 \text{ s}$

#Exercise 17

The open loop transfer function of a unity negative feed back system is $G(s) = \frac{K}{s(s+2)}$. A system response to a unit step input is specified as follows.

Peak time $t_p = 1.1s$,

Maximum overshoot=5%.

Determine whether both the specifications can be met simultaneously.

#Exercise 18

A unity negative feedback control system has the plant $G(s) = \frac{K}{s(s+\sqrt{2K})}$. Determine the percentage overshoot and 2% settling time for unit step input. For what range of K, the settling time is less than 1 s.

#Exercise 19

A closed loop control system has a transfer function $T(s) = \frac{500}{(s+10)(s^2+10s+50)}$. Plot its step response using Matlab/Scilab. Then consider the relatively dominant complex poles and plot its step response. Compare the results.

#Exercise 20

The control system for positioning the head of a cd drive has the closed loop transfer function $T(s) = \frac{0.313(s+0.8)}{(s+0.25)(s^2+0.3s+1)}$. Plot using Matlab/Scilab, the poles and zeroes and the step response of the system. What are the time response specifications like rise time, peak time, and overshoot. Now look at the dominance of the complex poles. If we consider the dominant poles only, how the transient response specifications change?